Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No: 6

Title: Creating HR Analytics Dashboards

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Creating HR Analytics Dashboards

Github Link: https://github.com/San-bytes/Data-Analytics-For-Business/tree/main/IBM-HR

Problem Statement:

You are an HR Data Analyst in a multinational organization. The HR department wants insights into employee demographics, hiring trends, attrition, and workforce distribution. Develop an HR analytics dashboard to support workforce planning and employee management.

Methodology

The following steps were followed to develop the HR Analytics Dashboard:

1. Imported the IBM HR Analytics Employee Attrition & Performance Dataset into Power BI.
2. Performed data cleaning and transformation using Power Query to ensure the data was accurate and ready for analysis.
3. Prepared and organized employee data based on demographics, job roles, departments, tenure, and attrition.
4. Created calculated measures using DAX to calculate key HR metrics such as total employees, attrition count, and attrition rate.
5. Designed an interactive dashboard using Power BI visuals such as:
    - KPI Cards
    - Tree map
    - Line Chart
    - Column Chart
    - Donut Chart
    - Bar Chart
    - Slicers
6. Added interactive slicers for Department, Business Travel, Education Field, and OverTime to enable detailed analysis.
7. Configured interactions between visuals so that KPIs and charts dynamically update based on selected filters.
8. Reviewed the dashboard to identify major attrition patterns and workforce trends that can support HR decision-making.

DAX Measures :

1. Average Monthly Rate:

- Average Monthly Rate=AVERAGE('IBM HR'[DailyRate])

```
Average Monthly Rate = AVERAGE('IBM HR'[DailyRate])
```

2. Total Employee:

- Total Employee=COUNT('IBM HR'[EmployeeNumber])

```
Total Employee = COUNT('IBM HR'[EmployeeNumber])
```

3. Total Attrition:

- Total Attrition=CALCULATE(COUNT('IBMHR'[EmployeeNumber]),'IBM HR'[Attrition] = "Yes")

```
Total Attriton = CALCULATE(COUNT('IBM HR'[EmployeeNumber]),'IBM HR'[Attrition]="Yes")
```

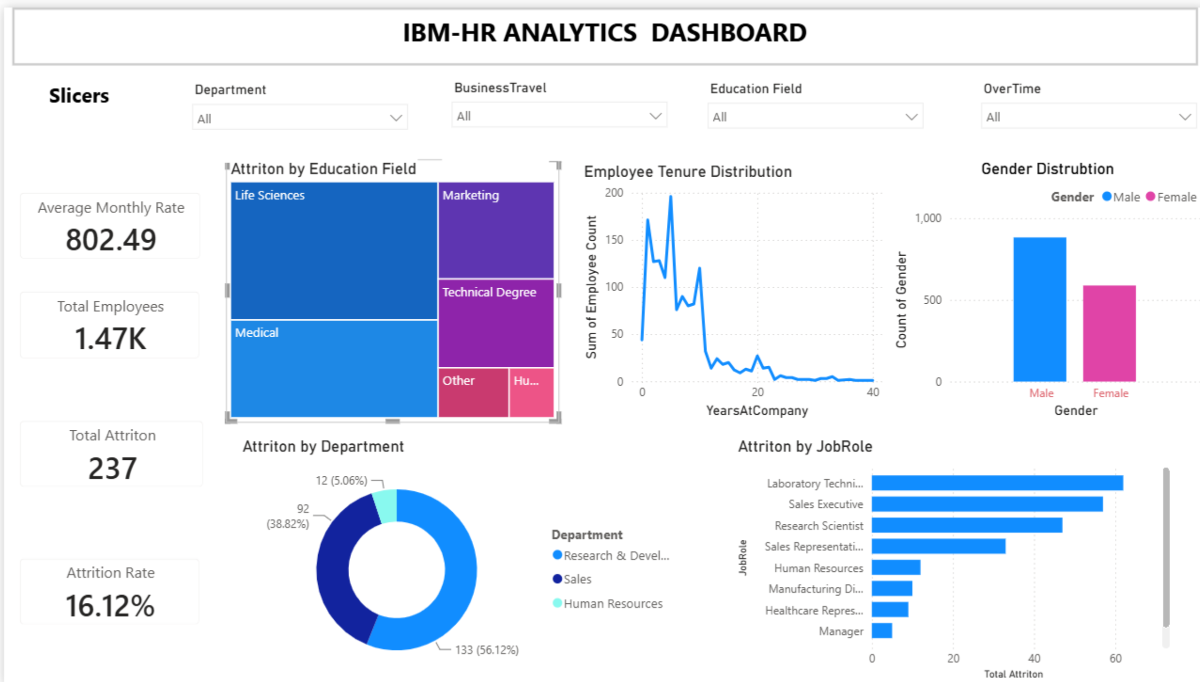
4. Attrition Rate:

- AttritionRate=DIVIDE([TotalAttrition],[Total Employee],0)

```
Attrition Rate = DIVIDE([Total Attriton],[Total Employee],0)
```

Dashboard Description

Dashboard Overview:

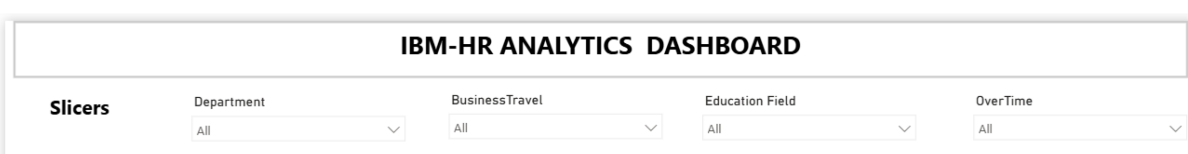


The HR Analytics Dashboard provides an interactive view of employee demographics, workforce distribution, employee tenure, and attrition patterns.

The dashboard contains four slicers:

- Department
- Business Travel
- Education Field
- Over Time

These slicers allow HR personnel to analyze specific employee groups and observe how the key metrics and charts change dynamically.

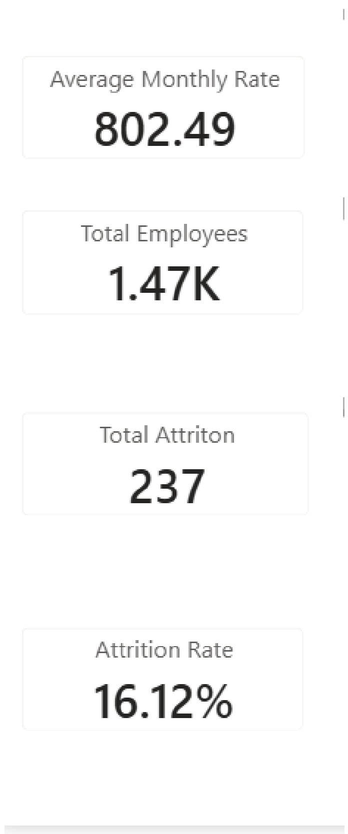


Key Performance Indicators:

The dashboard displays the following major KPIs:

- Average Monthly Rate: 802.49
- Total Employees: 1.47K
- Total Attrition: 237
- Attrition Rate: 16.12%

These KPIs provide a quick overview of the organization's workforce size and employee attrition.



Visualizations Used

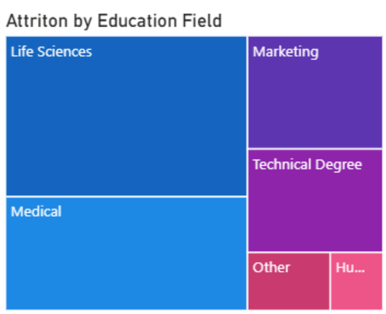
1. Attrition by Education Field

A Treemap is used to visualize attrition across different education fields.

The visualization includes categories such as:

- Life Sciences
- Medical
- Marketing
- Technical Degree
- Other
- Human Resources

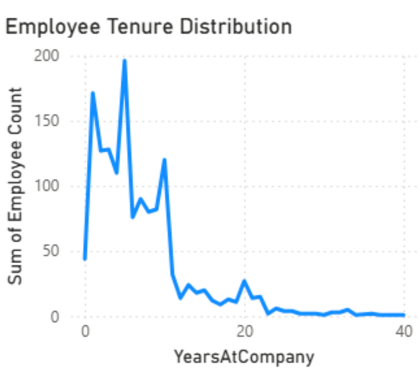
The size of each section represents the relative contribution of that education field to employee attrition.



2. Employee Tenure Distribution

A Line Chart is used to show the distribution of employees according to their Years at Company.

The chart helps identify how the workforce is distributed across different tenure levels. It shows that employee counts are generally higher during the early years of employment and gradually decrease as tenure increases.



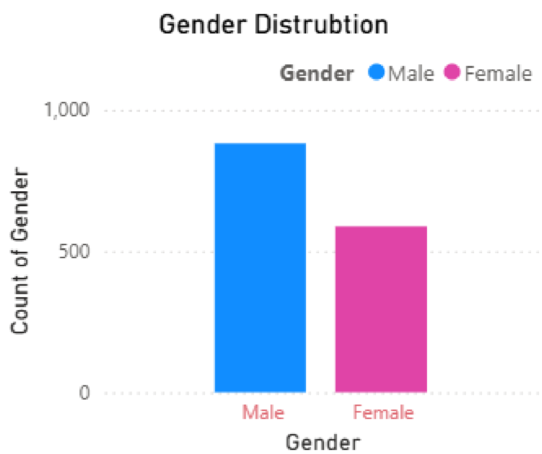
3. Gender Distribution

A Column Chart is used to compare the number of male and female employees.

The dashboard shows:

- Male employees: approximately 882
- Female employees: approximately 588

This visualization provides an overview of the gender composition of the workforce.

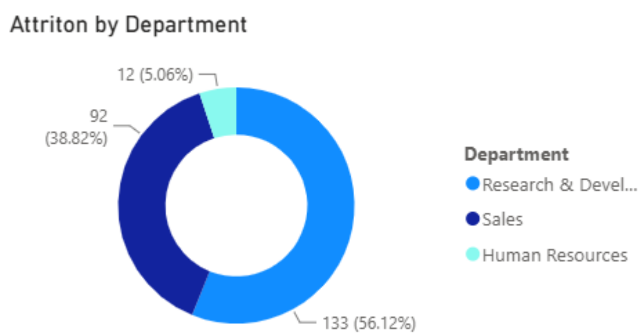


4. Attrition by Department

A Donut Chart is used to analyze employee attrition across different departments.

The dashboard mainly represents attrition across:

- Research & Development
- Sales
- Human Resources



This helps identify which departments contribute more significantly to overall employee attrition.

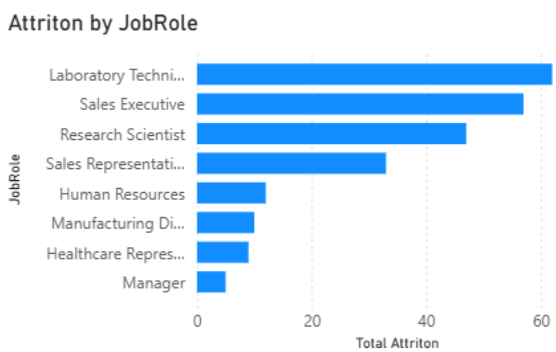
5. Attrition by Job Role

A Horizontal Bar Chart is used to compare attrition across different job roles.

The visualization shows higher attrition in roles such as:

- Laboratory Technician
- Sales Executive
- Research Scientist
- Sales Representative
- Human Resources
- Manufacturing Director
- Healthcare Representative
- Manager

This allows HR teams to identify job roles that may require greater attention from a retention perspective.



Interactive Filtering

The dashboard provides interactive filtering through four slicers:

1. **Department -** Allows HR users to analyze employees belonging to a particular department.
2. **Business Travel -** Allows comparison of employees based on their travel requirements, such as frequent travel.
3. **Education Field -** Allows analysis of workforce and attrition according to employees' educational backgrounds.
4. **Over Time -** Allows HR users to analyze employees based on whether they work overtime.
5. The dashboard automatically updates all KPIs and visualizations whenever a filter is applied.

Business Insights

- Overall attrition rate is 16.12%, with 237 employees leaving out of around 1.47K.
- Attrition is higher in roles such as Laboratory Technician, Sales Executive, and Research Scientist.
- Research & Development shows a significant share of employee attrition.
- Employees with fewer years of service form a large workforce segment, highlighting the need for early retention efforts.
- Male employees are higher in number than female employees.
- Life Sciences and Medical education fields contribute significantly to attrition.
- Slicers help analyze attrition by department, business travel, education field, and overtime.

Conclusion

The HR Analytics Dashboard provides a clear view of workforce demographics, tenure, and attrition using interactive Power BI visuals. With an overall attrition rate of 16.12%, the dashboard helps HR teams identify high-risk employee segments and supports data-driven workforce planning and retention decisions.